

## Project Design Phase-I

### Solution Architecture

|               |                                       |
|---------------|---------------------------------------|
| Date          | 19 June 2026                          |
| Team ID       | 2023011032040                         |
| Project Name  | AI-Based Threat Intelligence Platform |
| Maximum Marks | 4 Marks                               |

Solution Architecture:

The solution architecture covers the conceptual flow of how data moves from point A to point B within the AI-Based Threat Intelligence Platform.

#### Phase 1: Data Ingestion & Collection (The Inputs)

- Internal Sources: APIs and log collectors continuously gather data from the company's existing Firewalls, Endpoint Protection tools, and SIEMs.
- External Sources: Web scrapers and API connectors pull open-source threat feeds and Dark Web monitoring data.

#### Phase 2: Aggregation & Processing (The "Brain")

- Data Lake/Storage: All incoming structured and unstructured data is routed into a centralized cloud storage environment (e.g., AWS S3 or a similar data lake).
- AI/ML Correlation Engine: Machine learning algorithms pull from this storage to continuously monitor network traffic and identify anomalous behavior or indicators of compromise (IOCs).

#### Phase 3: Analysis & Decision Making (The Logic)

- Predictive Analytics Module: The system compares real-time anomalies against historical threat actor behavior to predict potential attack patterns.
- Risk Scoring: The platform assigns a severity score (e.g., 1-100) to the correlated data to prioritize alerts based on relevance.

#### Phase 4: Output & Action (The User Experience)

- Real-Time Alerts: The system triggers automated notifications (via email, SMS, or downstream IT systems) for critical threats.
- Security Analyst Dashboard: Processed intelligence is fed into an interactive, web-based UI where analysts can drill down into TTPs (Tactics, Techniques, and Procedures), view mitigation strategies, and perform ad-hoc queries.

